University of Antwerp
Algorithms and Complexity
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Exercises session 7

Reductions

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1 Definitions

Definition 1 (Graph colorability) *A graph $G = (V, E)$ is k-colorable if there's an assignment of k colours to vertices $c : V \to 1, \ldots, k$ such that for every $(u, v) \in E$, $c(u) \neq c(v)$*

Definition 2 (Planar graph) *A planar graph is a graph which **can** be drawn in the plane without any edges crossing*

2 Session exercises

Prove that the following problems are NP-complete

1. **3-COLORING**. Given $G = (V, E)$ decide if the graph is 3-colorable.
2. **PLANAR 3-COLORING**. Given $G = (V, E)$ decide if G is a planar 3-colorable graph.
3. **SET COVER**. Lets $\mathcal{U}$ be a set of elements, $\mathcal{S}$ a collection of subsets of $\mathcal{U}$ and k an integer. Decide if there's a collection of at most k subsets whose union is $\mathcal{U}$.
4. **KNAPSACK.** Given n items of sizes $s_1, \ldots, s_n$ and values $v_1, \ldots, v_n$, a capacity B and a value V, is there any subset $S \subseteq \{1, \ldots, n\}$ such that $\sum_{i \in S} s_i \leq B$ and $\sum_{i \in S} v_i \geq V$